DSA5205 Project 1: Machine Learning in Finance

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Abstract

This report studies model complexity, regularization, and out-of-sample performance in return prediction. Task 1 reproduces the misspecified ridge experiment of Kelly, Malamud, and Zhou (2024); the remaining sections document the extensions, hidden-data predictions, and financial interpretation of neural-network generalization.

1 Task 1: Misspecified Simulation with Ridge

1.1 Design and implementation

Following Kelly, Malamud, and Zhou [1], returns are generated as

$$R_{t+1} = S_t'\beta^* + \varepsilon_{t+1}, \qquad S_t \sim \mathcal{N}(0, I_P), \qquad \varepsilon_{t+1} \sim \mathcal{N}(0,1). \tag{1}$$

The vector β^* is drawn from an isotropic Gaussian distribution and normalized so that $\|\beta^*\|_2^2 = b^* = 0.2$. The data-generating complexity is fixed at $c = P/T_{\text{tr}} = 10$. In each replication, one random permutation of the P columns is held fixed across the full grid. For a given revealed fraction q, only the first $P_1 = qP$ permuted columns are observed, so observed complexity is $cq = P_1/T_{\text{tr}}$. This nested observation rule creates misspecification while preserving the sufficiently mixed setting.

The simulation parameters are listed in Table 1 and configured in `task1_2/task1.yaml`. Data generation is implemented by `generate_data()` in `task1_2/generate_data.py`; the experiment loop and nested feature selection are in `task1_2/task1.py`.

Table 1: Task 1 simulation parameters.

Parameter	Value
Base seed and replications	5205; 50
Training and test sizes	$T_{\text{tr}} = 500$; $T_{\text{te}} = 5000$
DGP complexity and dimension	$c = 10$; $P = 5000$
Signal strength and noise	$b^* = 0.2$; $\sigma_\varepsilon = 1$
cq grid	$\{0.50, 0.75, 0.90, 0.95, 0.98, 1.02, 1.05, 1.10, 1.25, 1.50, 2, 3, 5, 7, 10\}$
Ridge grid	$z \in \{0, 0.01, 0.1, 1, 10, 50\}$

For each observed block, ridge is estimated without an intercept:

$$\widehat{\beta}(z) = \left(X'X + zT_{\text{tr}}I\right)^{-1} X'y. \tag{2}$$

At $z = 0$, the implementation uses the Moore–Penrose pseudoinverse. For $z > 0$, it solves the primal system when $P_1 \le T_{\text{tr}}$ and the equivalent dual system otherwise; see `fit_ridge()` in `task1_2/fit_ridge.py`.

All quantities are evaluated on the independent test set. With $\widehat{R}_{t+1} = S_t'\widehat{\beta}$ and timing return $R^{\pi}_{t+1} = \widehat{R}_{t+1}R_{t+1}$, the reported metrics are

$$R^2_{\text{paper}} = 1 - \frac{\mathbb{E}[(R_{t+1} - \widehat{R}_{t+1})^2]}{\mathbb{E}[R^2_{t+1}]}, \tag{3}$$

$$\mathbb{E}[R^{\pi}_{t+1}] = \mathbb{E}[\widehat{R}_{t+1}R_{t+1}], \tag{4}$$

$$\text{SR} = \frac{\mathbb{E}[R^{\pi}_{t+1}]}{\sqrt{\mathbb{E}[(R^{\pi}_{t+1})^2]}}, \qquad \|\widehat{\beta}(z)\|_2. \tag{5}$$

Their sample analogues are computed by `calculate_metrics()` in `task1_2/test_metrics.py`.

1.2 Results

Figure 1 reports averages over the 50 replications. It is the finite-sample analogue of Figures 4–6 in Kelly, Malamud, and Zhou [1], which plot population limits rather than simulated estimates. The ridgeless specification displays the expected instability around the interpolation threshold $cq = 1$: R^2_{paper} falls to about -50 at $cq = 0.98$–1.02, while $\|\widehat{\beta}\|_2$ rises above 7.5. These extremes, and timing returns above 0.02, are clipped by the displayed axes. As in the paper's Figure 4, both effects abate for $cq > 1$ because overparameterized minimum-norm regression supplies implicit regularization. Explicit shrinkage smooths the interpolation singularity; $z = 10$ or 50 keeps R^2_{paper} close to zero and the coefficient norm small.

The expected-return panel has the same broad message as the left panel of the paper's Figure 5: very simple models have low timing returns because they observe too little of the DGP, and additional features improve approximation. In the population benchmark, ridgeless expected return rises to $b^*/c = 0.02$ at $cq = 1$ and is exactly flat thereafter. The simulation approaches roughly 0.018–0.019 away from interpolation, but overshoots and fluctuates near $cq = 1$; for example, its ridgeless mean is 0.0347 at $cq = 1.02$. This difference is consistent with finite training and test samples and the nonlinear effect of near-singular estimates, rather than a contradiction of the limiting result. The paper's Figure 5 also reports portfolio volatility; the project instead requires the Sharpe ratio and coefficient norm, which expose the same interpolation instability through risk adjustment and parameter dilation.

The Sharpe panel closely reproduces the paper's Figure 6. Ridgeless and lightly regularized curves show "double ascent": Sharpe rises at low complexity, dips near interpolation as coefficient and portfolio risk explode, and rises again for $cq > 1$. Nontrivial shrinkage removes most of this dip, producing an approximately permanent ascent. At $cq = 10$, all curves converge to about 0.049–0.050; the largest value on the simulated grid is 0.0505 for $z = 50$. For $\Psi = I$, the paper's optimal shrinkage is

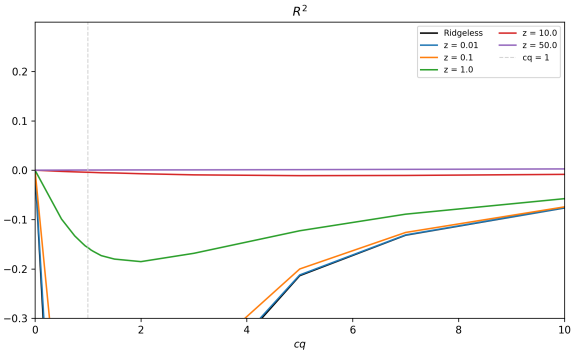
$$z^*(q) = \frac{c\{1 + b^*(1 - q)\}}{b^*} = 60 - 10q, \tag{6}$$

so $z = 50$ is exactly optimal at $q = 1$ and is close to the theoretical range over the grid. Theorem 1's monotone, concave improvement applies to this optimally adjusted $z^*(q)$, not to every fixed-z curve. Overall, the finite-sample experiment supports the virtue-of-complexity mechanism: under misspecification and sufficient mixing, observing more signals improves risk-adjusted timing performance even when conventional out-of-sample R^2 remains negative.

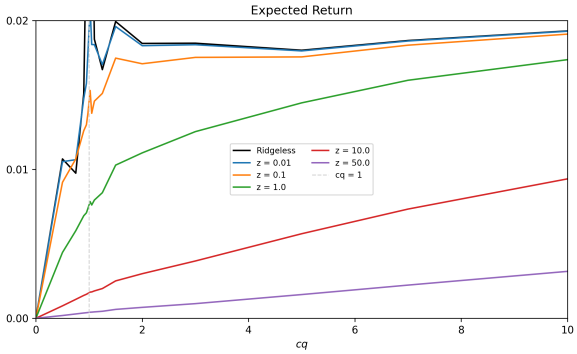
2 Task 2: Alternative Model

2.1 Method and experimental control

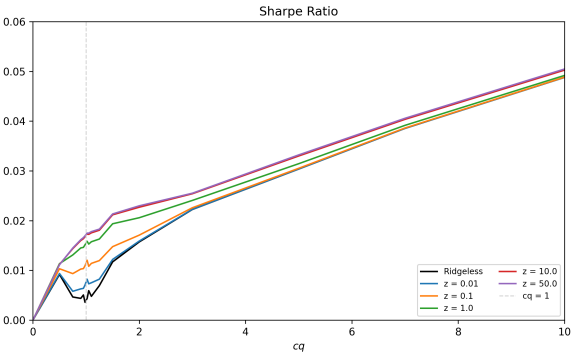
Task 2 is implemented in `task1_2/task2.py` and reads the shared configuration in `task1_2/task2.yaml`. For every replication it reuses the Task 1 seed, data-generating process, train/test



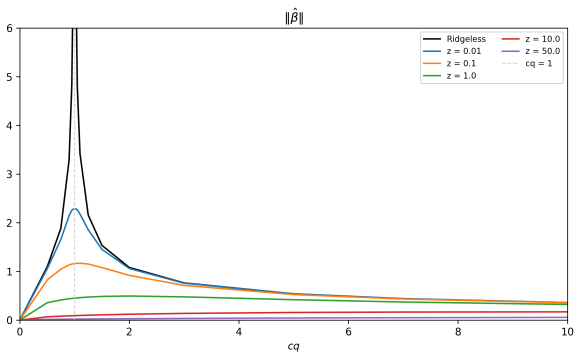
(a) R^2_{paper}.



(b) Expected timing return.



(c) Paper Sharpe ratio.



(d) Coefficient norm.

Figure 1: Finite-sample counterparts of Figures 4–6 in Kelly, Malamud, and Zhou [1], averaged over 50 replications. The dashed line marks $cq = 1$; displayed axes clip the interpolation extremes. All four panels are produced by `plot_results()` in `task1_2/plot_results.py` from `results.csv`.

split, fixed feature permutation, and cq grid, replacing ridge regression with Lasso regression. The estimator solves

$$\widehat{\beta}(\alpha) = \arg\min_{\beta} \left\{ \frac{1}{2T_{\text{tr}}} \|y - X\beta\|_2^2 + \alpha \|\beta\|_1 \right\}, \tag{7}$$

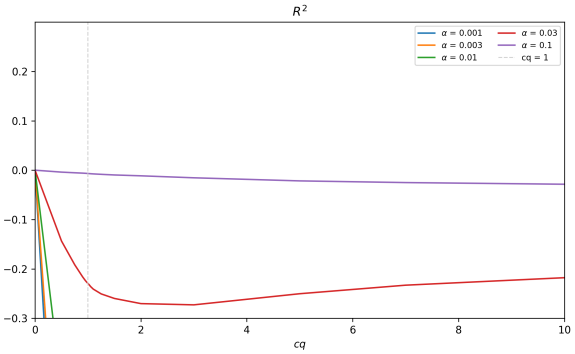
without an intercept. The fixed grid is $\alpha \in \{0.001, 0.003, 0.01, 0.03, 0.1\}$. The complete path is fitted by `lasso_path()` with warm starts, random coordinate selection, at most 3000 iterations, and tolerance 10^{-3}. The same out-of-sample metrics are then computed by `calculate_metrics()` in `task1_2/test_metrics.py`. Thus differences from Task 1 reflect the estimator rather than different simulated samples.

2.2 Results and comparison with ridge

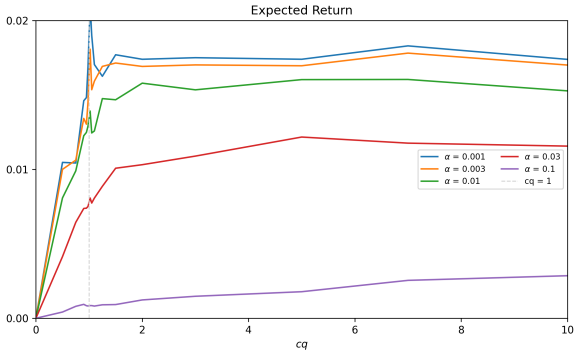
Figure 2 shows the averages over 50 replications. Weak penalties ($\alpha = 0.001$ and 0.003) retain an interpolation effect: near $cq = 1$, coefficient norms peak at approximately 2.54 and 1.71, and R^2_{paper} reaches about -5.36 and -2.41, respectively. This is substantially less extreme than ridgeless regression in Task 1, whose norm exceeds 7.5 and whose R^2_{paper} falls below -50, although it does not eliminate instability. Increasing α shrinks the norm and smooths the threshold. At $\alpha = 0.1$, R^2_{paper} remains near zero, but both expected return and Sharpe are small because the model is strongly underfit.

The economic metrics expose the bias–variance trade-off. The smallest penalty attains the largest expected timing return, 0.0211 at $cq = 1.02$, but its simultaneous prediction error keeps Sharpe at only 0.0075 there. For $cq > 1$, the weakly penalized Sharpe curves recover and reach about 0.0224 at $cq = 10$. Moderate $\alpha = 0.03$ gives the best Sharpe around the interpolation region and a relatively smooth curve, but it plateaus near 0.0197. Strong $\alpha = 0.1$ suppresses both signal and risk and reaches only 0.0129 at $cq = 10$.

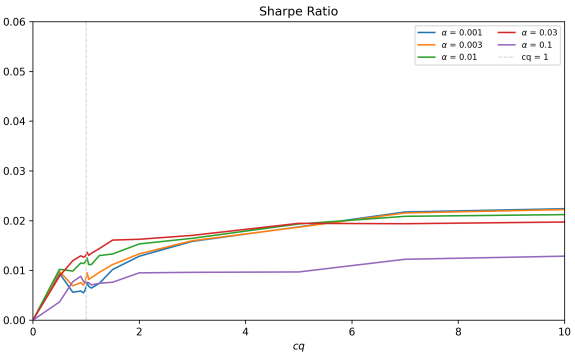
Lasso therefore reproduces the broad virtue-of-complexity pattern—timing performance generally improves as more signals are revealed—but it underperforms the best ridge specification on this dense Gaussian DGP. At $cq = 10$, the best Lasso Sharpe on the grid is 0.0224, compared with 0.0505 for ridge with $z = 50$; its corresponding R^2_{paper} is -0.390, compared with 0.0026 for ridge. This gap is consistent with model mismatch: the ℓ_1 penalty is designed to exploit sparsity, whereas the true β^* spreads a small signal across all P coordinates. Lasso discards many individually weak but collectively useful predictors, while the ℓ_2 penalty retains and stabilizes them.



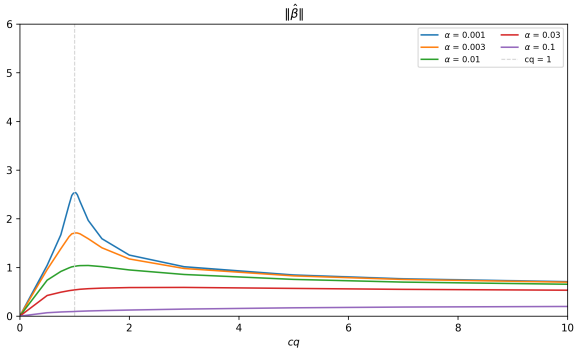
(a) R^2_{paper}.



(b) Expected timing return.



(c) Paper Sharpe ratio.



(d) Coefficient norm.

Figure 2: Out-of-sample Lasso results, averaged over 50 replications. The dashed line marks $cq = 1$ and the displayed axes clip some interpolation extremes. The panels are produced by `plot_lasso_results()` in `task1_2/plot_results.py` from `task1_2/task2_result_20261008_205656/results.csv`.

3 Task 3: Predictions for Data Sets A, B, and C

3.1 Validation and model selection

3.2 Final models and prediction files

4 Task 4: Generalization and Overfitting in Neural Networks

4.1 Main finding of Montanari and Urbani

4.2 Implication for financial machine learning

5 Task 5: Video and Reproducibility

6 Conclusion

References

[1] Kelly, B. T., Malamud, S., and Zhou, K. (2024). The virtue of complexity in return prediction. *Journal of Finance*, 79(1), 459–503.

[2] Montanari, A. and Urbani, P. (2025). Dynamical decoupling of generalization and overfitting in large two-layer networks. *Advances in Neural Information Processing Systems*, 38.